

The Ground Is Shifting

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Over the last nine years, I've worked across federal public health, academic research, and industry. My experience ranges from supporting NIAID and the Emory National Primate Research Center to building the national wastewater surveillance pipeline for SARS-CoV-2 and supporting several outbreak programs at CDC. More recently, I've productionized outbreak pipelines at Booz Allen and started building AI agentic applications for clinical use.

Across all these sectors, I have never seen the ground move under this profession the way it is moving right now. The shifts are happening simultaneously from multiple directions, and they are reinforcing each other in uncomfortable ways. This first issue is about understanding that terrain before we talk about what to do. **If you do not see the whole picture, you cannot make a good decision about where to stand.**

This is Issue 1 of a five-part series. In Issue 2 we go into data centers and energy. Issues 3 and 4 break down how the work is being redefined across every sector of life sciences. Issue 5 closes with the K-shaped economics of AI, the bubble debate, and a staged pivot framework.

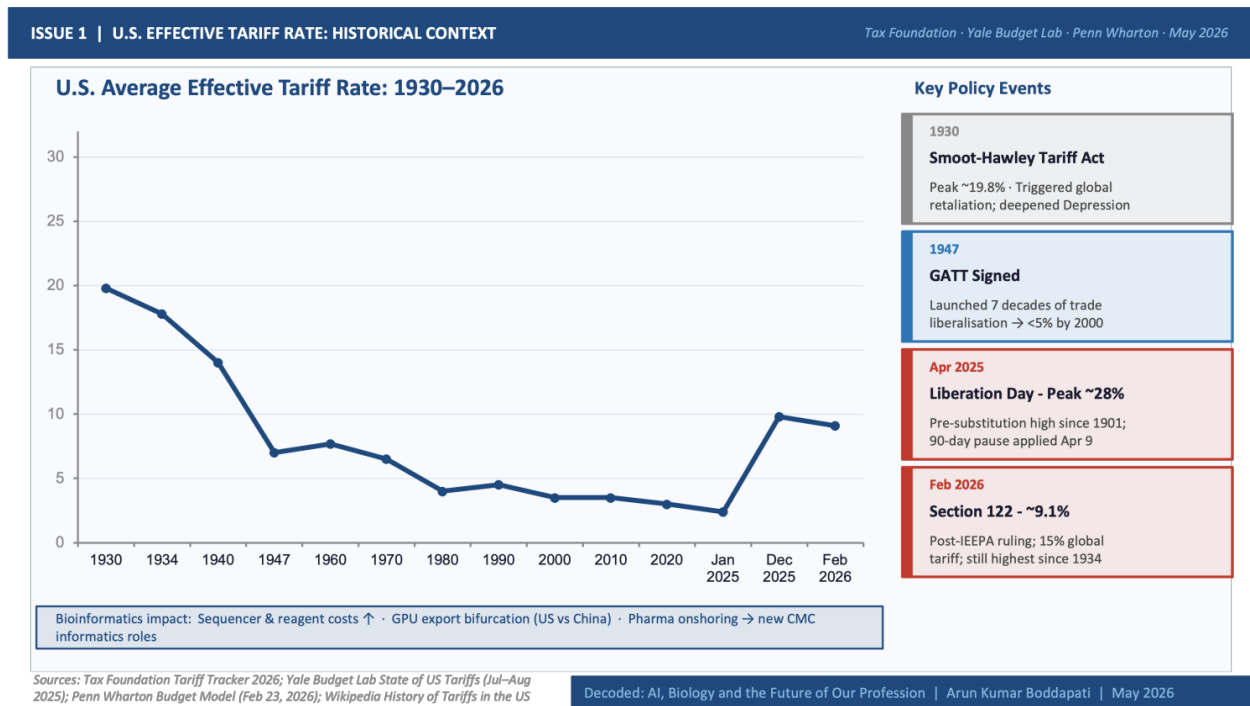
TL; DR: Five forces are converging on bioinformatics careers simultaneously.

- Tariffs and export controls are reshaping compute access and pharma supply chains.
- A hiring recession has stalled job growth across the economy, with biopharma shedding 22,000+ roles.
- Big Pharma is aggressively offshoring routine work to India and redirecting those savings into AI infrastructure and startup acquisitions.
- A \$100K H-1B fee and wage-weighted lottery are choking the international talent pipeline, accelerating offshoring further.
- Biological AI models like AlphaFold 3 have crossed a real performance threshold, justifying a \$600B+ annual capex buildout. Meanwhile, NIH funding is appropriated but barely flowing. The ground is shifting in every direction at once.

PART ONE: The Geopolitical and Trade Layer

The second Trump administration has adopted the most tariff-intensive economic posture in modern U.S. history. According to the Tax Foundation, the effective tariff rate has settled in the 9–11% range as of early 2026, the highest sustained level since the 1940s with active litigation over the legal authority for the 10% global tariff ongoing in the Court of International Trade [1].

On April 2, 2026, the administration issued a Section 232 executive order establishing a tiered tariff structure for patented pharmaceuticals, including a default 100% tariff for most countries and reduced rates of 15–20% for allied nations and companies with approved U.S. onshoring initiatives. Generics, biosimilars, and orphan drugs remain exempt, with implementation for large manufacturers beginning July 31, 2026 [1].



In parallel, semiconductors are under a separate Section 232 investigation. On January 14, 2026, the administration imposed a 25% tariff on advanced AI chips intended for re-export to China, specifically targeting products such as Nvidia H200 and AMD MI325X accelerators while exempting chips deployed within U.S.-based data centers [2].

For bioinformatics professionals, three transmission channels matter most. First, reagent and instrument supply chains (such as sequencers, library prep kits, and cold-chain biologics) face elevated landed costs being absorbed into pharma cost-cutting programs. Second, semiconductor export controls are bifurcating the GPU landscape. US-based biological AI teams get preferred access to Blackwell-class silicon, whereas Chinese labs are being pushed into domestically optimized stacks. This matters for how foundation models in biology will be trained globally. Third, pharmaceutical onshoring announcements from Lilly, Pfizer, Novartis, and Roche are creating new computational roles in process analytics, digital twins, and CMC informatics, a career path that barely existed five years ago.

On Russia-Ukraine, now in its fifth year, and Middle East escalations through 2025/6, the defense supply chain is stretched and the US fiscal posture is constrained, adding political pressure to an already contentious federal spending environment. But while capital and hardware are being reshuffled by trade policy, the most immediate shock to our daily work is happening on the human side of the equation.

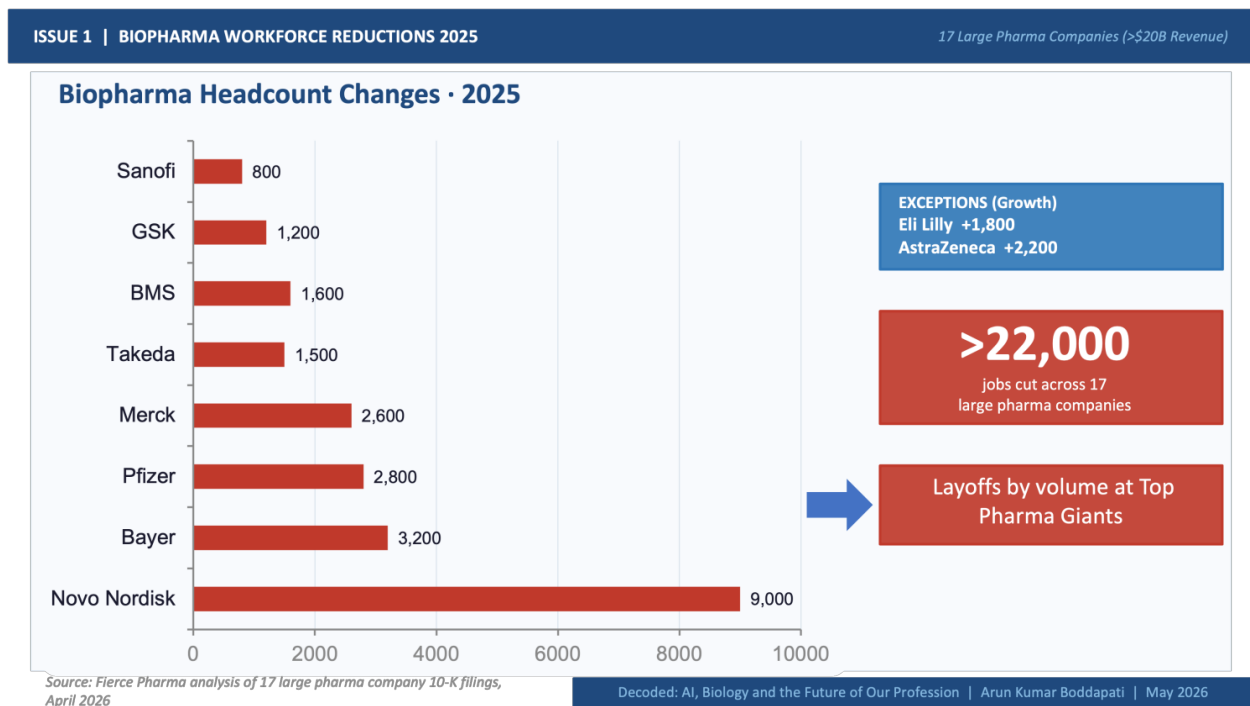
PART TWO: The Talent Pipeline Under Pressure

This is the part of the story that hits our profession most directly, regardless of your citizenship status or career stage.

The Hiring Recession: The US added approximately 181,000 jobs in calendar 2025 after the January 2026 benchmark revision. This represents the **weakest annual payroll growth in decades**, and a figure that is almost entirely attributable to healthcare and social services. Strip those sectors out, and the broader economy actually shrank its workforce in 2025. That kind of contraction is historically reserved for recessions [7,8].

Heather Long, chief economist at Navy Federal Credit Union and former Washington Post economics correspondent, coined the framing that captured it best. She called it a “hiring recession” or “jobless boom” to describe growth that looks strong on paper even while the hiring machine largely stopped [6]. Unemployment moved from **4.0% in January 2025 to 4.4% by December and then back to 4.3%, maintaining steady past couple of months.**

In biopharma specifically, large pharmaceutical companies have executed significant workforce reductions as they brace for a changing economic landscape [9].



Throughout late 2025, companies like Bayer, BMS, Merck, Pfizer, Takeda, GSK, and Sanofi all announced sweeping restructurings and layoffs. The notable exceptions were Lilly, AstraZeneca, and Amgen, which continued growing. The pattern is not random; it tracks closely with pipeline performance relative to the looming patent cliff and competitive pressure from AI-native drug discovery companies.

The Strategic Reallocation: The India Offshoring Accelerant

This restructuring is not just a reduction in force; it is a geographic and strategic pivot. As biopharma companies brace for the patent cliff, they are aggressively offshoring routine bioinformatics pipelines and early-stage discovery work to Global Capability Centers (GCCs) and CROs in India. The scale of this shift is staggering.

As of 2025, over 80 healthcare and life sciences GCCs operate in India, employing more than 280,000 professionals. Roughly half of the top 50 global pharma companies now have GCCs in the country, many established within the last five years [16]. These are not back-office call centers if you are wondering, they are fully capable offshore units. Indian GCCs now manage an estimated 45% of global drug discovery and development workflows, 60% of regulatory affairs, and 54% of pharmacovigilance for their parent organizations [16].

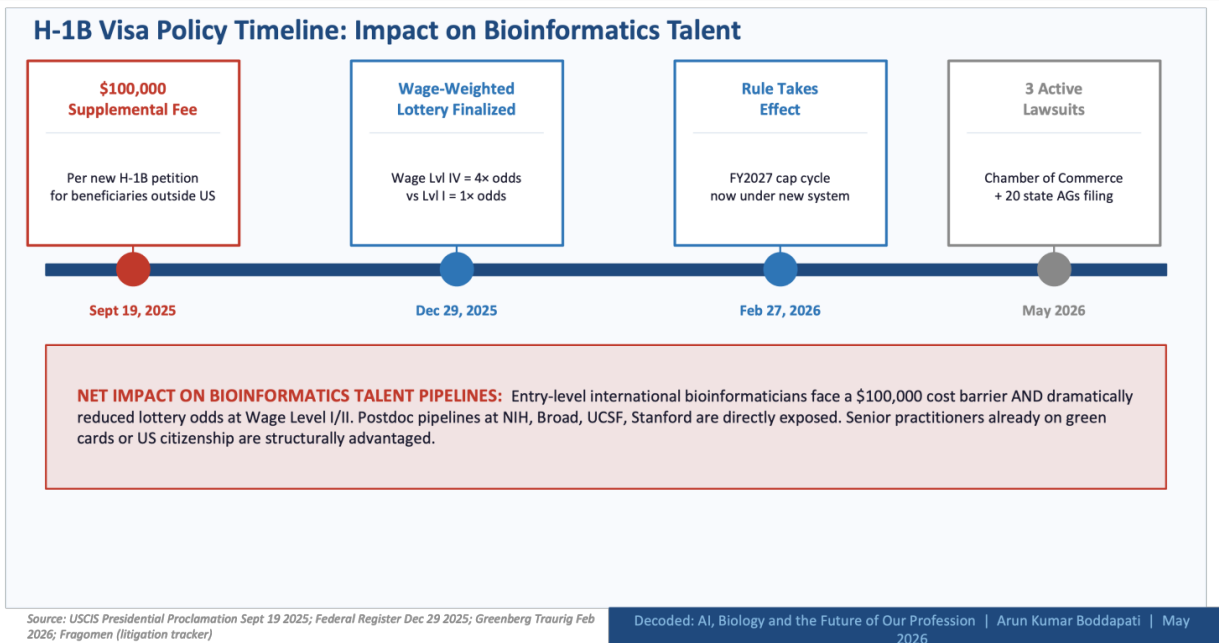
The cost math is straightforward: GCC setups deliver 40–50% average cost savings over equivalent US or European operations, with R&D, IT, and bioinformatics functions averaging roughly 40% savings [17]. India's pharma exports are nearing \$30 billion, fueled in part by this CRO/CDMO/GCC ecosystem [18]. The capital saved from this offshoring is being directly reallocated to fund the massive AI infrastructure and startup acquisitions discussed in Part Three. For the US-based professional, this means "routine" is no longer a safe career category; it is now an offshore-able commodity. **More on which roles fall on which side of that line in Issue 3.**

The H-1B Shock: Why This Accelerates Offshoring

There is a compounding effect here that most people miss. When companies cannot bring international talent to the US, they do not simply hire domestically instead. They offshore the function entirely. The H-1B crackdown is not just an immigration story; it is an accelerant for the India offshoring trend above.

On September 19, 2025, a Presidential Proclamation imposed a \$100,000 supplemental fee on new H-1B petitions for beneficiaries outside the United States [3]. The fee applies to new petitions and many change-of-status filings; it does not apply to extensions with the same employer. Three lawsuits have been filed challenging the proclamation by the US Chamber of Commerce, Global Nurse Force, and a coalition of 20 state AGs [5]. **As of May 2026, the fee remains in effect.**

Furthermore, the administration has directed USCIS to pursue a wage-weighted H-1B cap selection rule to prioritize high-skilled, high-paid workers over entry-level applicants [4]. Under such a system, registrations at Wage Level IV would receive heavy priority over Level I. For entry-level bioinformatics analysts and postdocs (historically Level I and II roles), the looming threat of a wage-weighted lottery means dramatically reduced odds-on top of the **\$100,000 cost barrier**.



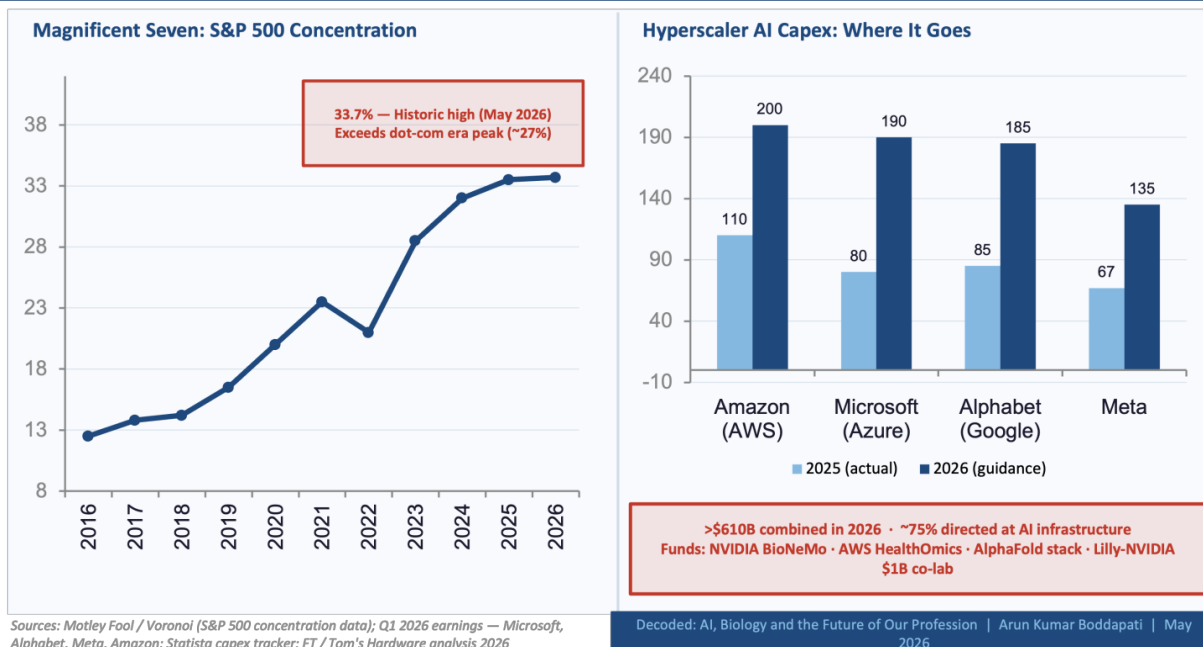
The life sciences workforce is disproportionately international. Postdoc pipelines at the Broad, MSK, UCSF, Stanford, and most academic medical centers are heavily foreign-born.

The new H-1B economics will compress junior international hiring in industry and raise the effective salary floor for new H-1B sponsorship to Wage Level III or above. It structurally advantages senior practitioners already holding green cards or US citizenship. This includes mid-career professionals with the kind of publications and domain depth that qualify for EB-1A or EB-2 NIW pathways. If that describes you, the immigration calculus has shifted materially in your favor. If you are currently H-1B-dependent and early-career, you need to be planning your path to permanent residency with urgency.

PART THREE: The AI Capex Signal

The Magnificent Seven (Apple, Microsoft, Alphabet, Amazon, Nvidia, Meta, and Tesla) now represent roughly **33.7% of the S&P 500** by market cap, their highest concentration in modern history. This isn't just a portfolio-management quirk. It is a massive, flashing signal about where capital is flowing, and that direction is overwhelmingly into AI infrastructure [10].

Microsoft, Alphabet, Meta, and Amazon together are projected to spend over \$610 billion on capital expenditure in 2026, representing a massive jump from 2025's already-record levels [11]. Roughly 75% of that is directed at AI: data centers, GPUs, networking, and power infrastructure. This is the funding source for the tools our field will use in 2026, 2027, and 2028.



How is the Money moving now?

This \$600B+ spend is not just speculative hype but a reaction to a concrete inflection in technical performance. Between 2024 and early 2026, biological AI models crossed the threshold from research curiosities to production-viable tools with measurable advantages over traditional methods.

The clearest signal is AlphaFold 3. On the PoseBusters benchmark, AF3 is roughly 50% more accurate than the best traditional physics-based docking tools (like Schrodinger) for predicting protein-ligand interactions and it requires no prior structural input to do so [19]. On protein-ligand docking (FoldBench), AF3 achieves a 64.9% success rate, substantially outperforming classical docking software and even other AI models like Boltz-1. For covalent ligand prediction, AF3 achieves a 98.3% AUC for distinguishing binders from decoys, "dramatically outperforming" classical tools [20]. Such similar positive data points are driving the Capex signal. When an AI model consistently outperforms decades of physics-based simulation on the benchmarks that matter most for drug design, the economics of the entire discovery pipeline change. The market is not betting on a future promise; it is reacting to a demonstrated, peer-reviewed capability inflection [21].

Nvidia BioNeMo, AWS HealthOmics, Google's AlphaFold stack, and the NVIDIA-Lilly co-innovation lab all sit on top of this buildout. What gets built in the next 24 months determines what the biological AI toolkit looks like for the next decade.

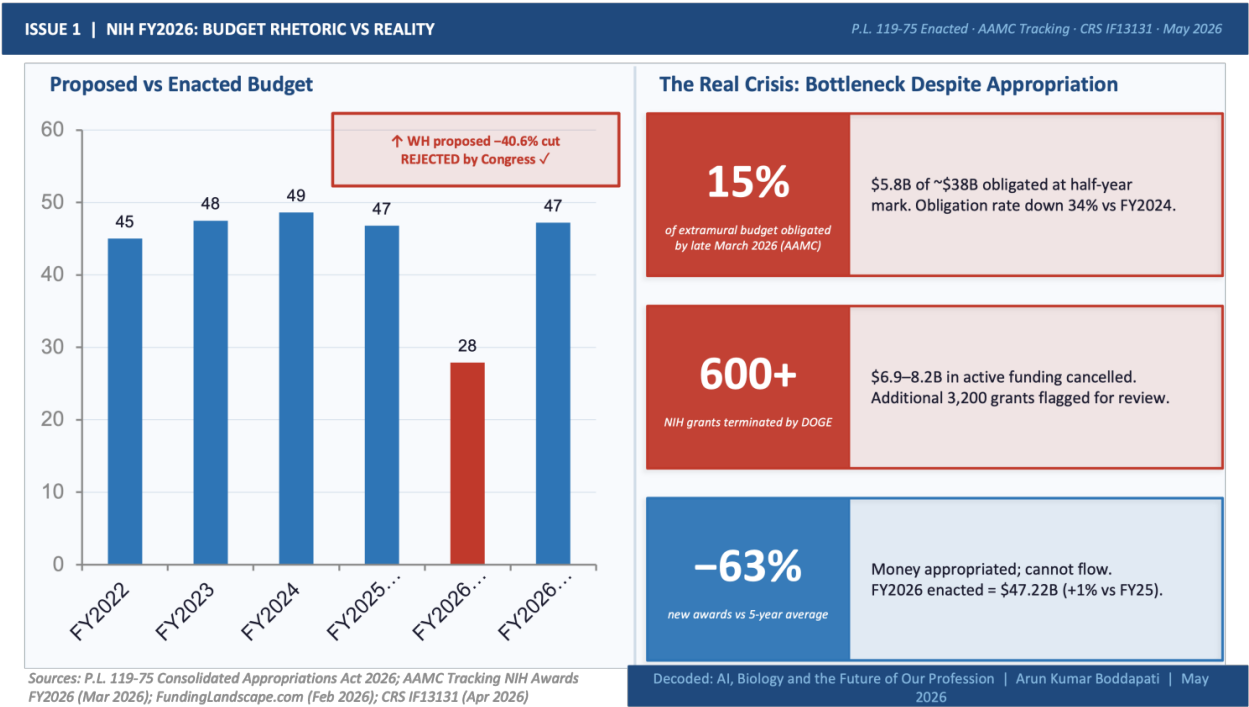
There is genuine debate about whether this capex is justified by the revenue it will generate. I'll cover that argument in full in Issue 2 (data centers and compute) and Issue 5 (the bubble debate). The short version for now is that the revenue gap is large, and the enterprise ROI is modest so far, but the infrastructure is being built regardless. For practitioners, what matters is that the tools

are coming whether the economics close in the short term or not. The question is whether you are positioned to use them today and an AI future ahead.

PART FOUR: The Federal Bottleneck and the Academic Squeeze

This part of the story is the most consequential for practitioners who trained in or work adjacent to the government-funded research ecosystem. For much of late 2025, the academic world braced for an existential shock, and while the nature of the shock changed, the impact remains severe.

The second Trump administration’s initial FY2026 budget proposed sweeping structural changes: a 40.6% reduction to the NIH budget (down to \$27.5 billion), consolidating 27 institutes into 8, massive HHS reductions in force, and a devastating 15% cap on indirect costs [12, 13].



But Congress rejected the vast majority of this. The final FY2026 budget provided a slight increase to NIH R&D funding (approving \$47.2 billion), explicitly blocked the indirect cost caps, and prevented the sweeping agency reductions from materializing [14].

However, the sustained threat of these cuts, combined with a post-government shutdown hangover and administrative policy changes, has created a massive funding bottleneck. As of May 2026, the NIH had obligated roughly 15% of its extramural budget, an incredibly slow pace for halfway through the fiscal year. **Overall obligation rates are down 34% compared to FY2024, and new awards have plummeted by 63% against the five-year average [15].**

The Academic Cascade: Even though the money was eventually appropriated, the cash-flow freeze triggered real-world panic. The Broad Institute laid off 75 employees in mid-2025, explicitly attributing the cuts to anticipated NIH funding reductions. UMass Chan Medical School

furloughed staff, and institutions like Northwestern, Columbia, and Washington State University implemented hiring freezes to protect their burn rates against the uncertainty.

The out-migration from academic bioinformatics to industry is accelerating as a result of this bottleneck. The most exposed roles include postdocs dependent on H-1B (now effectively blocked for new entrants at Level I wages), staff scientists paid from indirect cost recovery, and core facility analysts whose positions rely on steady grant cash flow. I am watching several colleagues navigate this in real time. The exit is not always comfortable, but given the friction in federal grant execution, it is increasingly the rational decision.

What This Means Together

Connecting all the threads together, we get a tightening immigration policy that's compressing the talent supply at exactly the moment when big pharma is offshoring routine computational work to India, freeing up billions in capital. That capital is being poured into an AI infrastructure buildout justified by a real change: **foundation models like AlphaFold 3 are now measurably outperforming decades of physics-based drug design tools.** Meanwhile, NIH grant execution is severely bottle-necked, pushing academic talent into an industry market that is simultaneously shedding entry-level roles and paying a premium for senior practitioners who can direct and validate these new AI systems.

The ground is not shifting in just one direction. It is shifting simultaneously toward offshore commoditization of routine work, toward AI-driven automation of mid-level analysis, and toward acute scarcity of domain-deep professionals who can bridge biology, production engineering, and regulatory validation. Where you sit on that spectrum determines whether this moment is a threat or the opportunity of your career.

NEXT ISSUE: Issue 2 covers the AI infrastructure build out, energy constraints, nuclear power deals, compute scarcity, and what it means for biological foundation model training. The data center problem is the most under reported story in all of AI.

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The Energy Problem

By Arun Kumar Boddapati | *Bioinformatics & Clinical Genomics* | NIH/NIAID (FNLRC) · Ex-Emory · Ex-CDC (Leidos, Booz Allen Hamilton) | *AI in Life Sciences* | May 2026

SERIES CONTEXT In Issue 1, I mapped the geopolitical, immigration, and federal funding pressures reshaping the bioinformatics operating environment. This issue goes one layer deeper into the physical infrastructure that will determine whether the AI buildout can actually deliver what it promises.

TL; DR: Four physical and economic realities are creating an immediate bottleneck for AI.

- The biggest constraint is no longer algorithms or data, but electricity as global data centers are on track to double their 415 TWh consumption by 2030, with GPUs accounting for only 40% of the massive facility power draw.
 - A severe physical supply chain failure has stretched transformer lead times to 3–5 years, delaying or cancelling nearly half of planned 2026 data center builds.
 - In response, tech giants are aggressively locking up the energy supply by signing 20-year nuclear power purchase agreements and acquiring energy developers outright.
 - For bioinformatics teams, this physical reality translates directly into severe GPU pricing volatility and the rise of a critical new premium skillset: Compute Sovereignty.
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There is a story playing out right now that most people in AI and data infrastructure are underestimating, but it may be the single most important constraint on the tools we use over the next decade. It has nothing to do with model architectures, training tricks, or bigger datasets, it's about Power-specifically, whether energy grids can physically supply the energy that large-scale AI will demand in the near future.

I want to be direct and upfront, I am not an Infrastructure Engineer, but as someone who has spent the better part of a decade running large-scale genomics workflows on AWS and HPC systems building pipelines that depend on cloud computing, this story affects me and everyone in this field. The question of whether GPU capacity is available, how it is priced, whether it's stable